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COMPUTER NETWORKS

Ex.No.1	**Study of various Network commands used in Linux and Windows: Hands-on practice of various network commands.**
Date:	

AIM: - To Study of various Network commands used in Linux and Windows

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	`arp -a`	`arp -a`	Displays the ARP cache, showing the IP addresses and corresponding MAC addresses of devices on the local network.	`arp -a`	`Interface: 192.168.1.10 --- 0x7 Internet Address Physical Address Type 192.168.1.1 00-1a-2b-3c-4d-5e dynamic`
2	`hostname`	`hostname`	Displays the hostname (computer name) of the local system.	`hostname`	`DESKTOP-PC01`
3	`ipconfig /all`	`ipconfig /all`	Displays complete TCP/IP configuration, including IP address, subnet mask, gateway, DNS, DHCP, and MAC address.	`ipconfig /all`	`IPv4 Address : 192.168.1.10 Default Gateway : 192.168.1.1 DNS Server : 8.8.8.8 DHCP Enabled : Yes`
4	`nbtstat -a`	`nbtstat -a <hostname>`	Displays NetBIOS name table of a remote computer for troubleshooting	`nbtstat -a DESKTOP-PC01`	`NetBIOS Remote Machine Name Table Name Type Status DESKTOP-PC01 <00> UNIQUE Registered`

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			NetBIOS over TCP/IP.		
5	`netstat`	`netstat`	Displays active TCP/IP connections, listening ports, routing tables, and interface statistics.	`netstat`	`Proto Local Address Foreign Address State TCP 192.168.1.10:49762 142.250.190.78:443 ESTABLISHED`
6	`netstat -r`	`netstat -r`	Displays the IP routing table of the local computer.	`netstat -r`	`IPv4 Route Table Network Destination Netmask Gateway Interface`
7	`nslookup`	`nslookup <domain>`	Queries the DNS server to obtain the IP address or DNS records of a domain.	`nslookup www.google.com`	`Server: dns.google Address: 8.8.8.8 Name: www.google.com Address: 142.250.190.68`
8	`pathping`	`pathping <hostname/IP>`	Combines the features of `ping` and `tracert` to trace the route and measure packet loss at each hop.	`pathping www.google.com`	`Tracing route to www.google.com Computing statistics... Hop 1 192.168.1.1 Hop 2 10.0.0.1 Packet Loss: 0%`
9	`ping`	`ping <hostname/IP>`	Tests connectivity between the local computer and another host using ICMP Echo Request and Reply messages.	`ping www.facebook.com`	`Pinging www.facebook.com [157.240.22.35] with 32 bytes of data: Reply from 157.240.22.35: bytes=32 time=25ms TTL=54`
10	`route`	`route print`	Displays or modifies the IP routing table. It is commonly used to view or configure static routes.	`route print`	`IPv4 Route Table Active Routes: Network Destination Netmask Gateway Interface`

Some important Linux networking commands

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	ip	`ip <OPTIONS> <OBJECT> <COMMAND>`	Displays and manages IP addresses, routing tables, and network interfaces.	`ip address show`	`2: eth0: <UP,BROADCAST,RUNNING> inet 192.168.1.100/24 brd 192.168.1.255 scope global eth0`
2	ip address add	`ip address add <IP>/<Prefix> dev <Interface>`	Assigns an IP address to a network interface.	`ip address add 192.168.1.254/24 dev eth0`	`Assigned 192.168.1.254/24 to eth0 successfully.`
3	ip address del	`ip address del <IP>/<Prefix> dev <Interface>`	Removes an IP address from a network interface.	`ip address del 192.168.1.254/24 dev eth0`	`Deleted 192.168.1.254/24 from eth0.`
4	ip link set up	`ip link set <Interface> up`	Brings the network interface online.	`ip link set eth0 up`	`Interface eth0 is UP.`
5	ip link set down	`ip link set <Interface> down`	Brings the network interface offline.	`ip link set eth0 down`	`Interface eth0 is DOWN.`
6	ip link set promisc on	`ip link set <Interface> promisc on`	Enables promiscuous mode on the interface.	`ip link set eth0 promisc on`	`Promiscuous mode enabled on eth0.`
7	ip route add default	`ip route add default via <Gateway> dev <Interface>`	Adds the default gateway for all outgoing traffic.	`ip route add default via 192.168.1.254 dev eth0`	`Default route added successfully.`
8	ip route add	`ip route add <Network> via <Gateway>`	Adds a static route through a gateway.	`ip route add 192.168.1.0/24 via 192.168.1.254`	`Route added successfully.`

S.No	Command	Synopsis	Description	Sample Input	Sample Output
9	ip route add (Direct Route)	`ip route add <Network> dev <Interface>`	Adds a direct route through a network interface.	`ip route add 192.168.1.0/24 dev eth0`	`Route added via eth0.`
10	ip route delete	`ip route delete <Network> via <Gateway>`	Deletes a static route.	`ip route delete 192.168.1.0/24 via 192.168.1.254`	`Route deleted successfully.`
11	ip route get	`ip route get <Destination IP>`	Displays the route used to reach a destination.	`ip route get 10.10.1.4`	`10.10.1.4 via 192.168.1.254 dev eth0 src 192.168.1.100`
12	ifconfig	`ifconfig`	Displays and configures network interface parameters (deprecated in modern Linux).	`ifconfig`	`eth0: flags=4163<UP,BROADCAST,RUNNING> inet 192.168.1.100 netmask 255.255.255.0`
13	mtr	`mtr <hostname/IP>`	Combines Ping and Traceroute to analyze network paths.	`mtr google.com`	`HOST: localhost Loss% Snt Last Avg Best Wrst StDev`
14	mtr -g	`mtr -g <hostname>`	Displays numeric IP addresses instead of hostnames.	`mtr -g google.com`	`192.168.1.1 → 142.250.190.78`
15	mtr -b	`mtr -b <hostname>`	Displays both hostnames and IP addresses.	`mtr -b google.com`	`router.local (192.168.1.1) google.com (142.250.190.78)`
16	mtr -c	`mtr -c <count> <hostname>`	Sends a specified number of probes.	`mtr -c 10 google.com`	`10 probes completed successfully.`
17	tcpdump -D	`tcpdump -D`	Lists all available	`tcpdump -D`	`1.eth0 2.lo 3.wlan0`

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			network interfaces.		
18	tcpdump	tcpdump -i <Interface>	Captures packets on a network interface.	tcpdump -i eth0	Listening on eth0...
19	tcpdump -c	tcpdump -i <Interface> -c <count>	Captures a specified number of packets.	tcpdump -i eth0 -c 10	10 packets captured.
20	tcpdump host	tcpdump -i <Interface> host <IP>	Captures traffic to and from a specific host.	tcpdump -i eth0 host 8.8.8.8	Captured packets to/from 8.8.8.8.
21	tcpdump src host	tcpdump -i <Interface> src host <IP>	Captures packets originating from a specific host.	tcpdump -i eth0 src host 8.8.8.8	Source packets from 8.8.8.8 displayed.
22	tcpdump dst host	tcpdump -i <Interface> dst host <IP>	Captures packets destined for a specific host.	tcpdump -i eth0 dst host 8.8.8.8	Destination packets to 8.8.8.8 displayed.
23	tcpdump net	tcpdump -i <Interface> net <Network>/<Prefix>	Captures traffic from a specified network.	tcpdump -i eth0 net 10.1.0.0/24	Packets from network 10.1.0.0/24 displayed.
24	tcpdump port	tcpdump -i <Interface> port <Port>	Captures traffic on a specific port.	tcpdump -i eth0 port 53	DNS packets captured.
25	tcpdump host and port	tcpdump -i <Interface> host <IP> and port <Port>	Captures traffic for a specific host and port.	tcpdump -i eth0 host 8.8.8.8 and port 53	DNS traffic with host 8.8.8.8 displayed.
26	tcpdump port not	tcpdump -i <Interface> port not <Port1> and not <Port2>	Captures all traffic except the specified ports.	tcpdump -i eth0 port not 53 and not 25	Traffic excluding ports 53 and 25 captured.
27	ping	ping <hostname/IP>	Tests network connectivity using ICMP	ping google.com	64 bytes from google.com: icmp_seq=1 ttl=117 time=10.5 ms

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			Echo Requests and Echo Replies.		
28	ping -c	`ping -c <count> <hostname>`	Sends a specified number of ICMP Echo Requests.	`ping -c 10 google.com`	`10 packets transmitted, 10 received, 0% packet loss.`

Configuring an Ethernet connection by using nmcli:

Step No.	Task	Command	Expected Result
1	List all Network Manager connection profiles	`nmcli connection show`	Displays all available network connection profiles with their **Name, UUID, Type, and Device**.
2	Create a new Ethernet connection profile It is specifically an **`nmcli` connection management command**. • **`nmcli`** → NetworkManager Command Line Interface • **`connection`** → Manages network connection profiles. • **`add`** → Creates a new connection profile.	`nmcli connection add con-name Internal-LAN ifname enp1s0 type ethernet`	Creates a new Ethernet connection profile named **Internal-LAN**.
3	Rename an existing connection profile (Optional)	`nmcli connection modify "Wired connection 1" connection.id Internal-LAN`	Renames the existing connection profile to **Internal-LAN**.

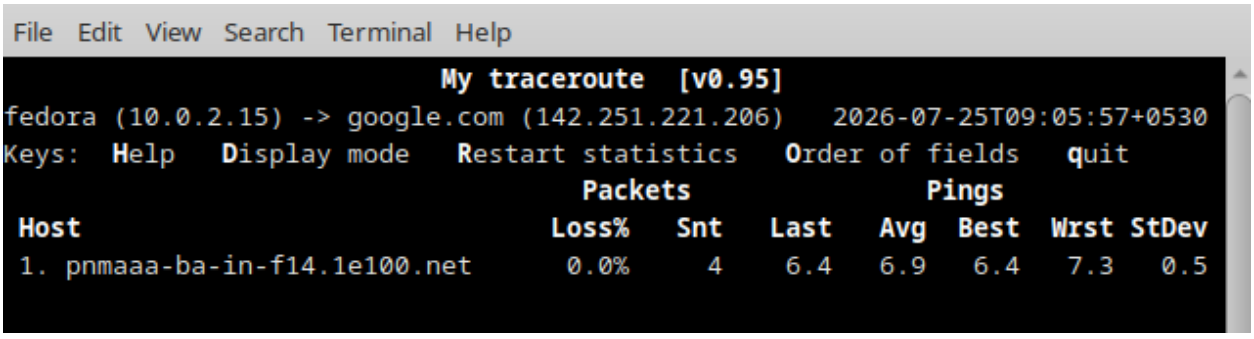
Step No.	Task	Command	Expected Result
4	Display the current settings of the connection profile	`nmcli connection show "Internal-LAN"`	Displays interface name, autoconnect status, IPv4 and IPv6 configuration details.
5	Configure IPv4 using DHCP	`nmcli connection modify "Internal-LAN" ipv4.method auto`	Configures the interface to obtain the IPv4 address automatically using DHCP.
6	Configure a static IPv4 address	`nmcli connection modify "Internal-LAN" ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com`	Assigns a static IPv4 address, gateway, DNS server, and DNS search domain.
7	Configure IPv6 using SLAAC	`nmcli connection modify "Internal-LAN" ipv6.method auto`	Enables automatic IPv6 configuration using Stateless Address Autoconfiguration (SLAAC).
8	Configure a static IPv6 address	`nmcli connection modify "Internal-LAN" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::ffee ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com`	Assigns a static IPv6 address, gateway, DNS server, and search domain.
9	Activate the connection profile	`nmcli connection up Internal-LAN`	Activates the **Internal-LAN** network connection profile.
10	Verify the IP address configuration	`ip address show enp1s0`	Displays the assigned IPv4 and IPv6 addresses of the **enp1s0** network interface.
11	Verify the IPv4 default gateway	`ip route show default`	Displays the configured IPv4 default gateway.
12	Verify the IPv6 default gateway	`ip -6 route show default`	Displays the configured IPv6 default gateway.

Step No.	Task	Command	Expected Result
13	Verify the DNS configuration	`cat /etc/resolv.conf`	Displays the configured DNS servers and DNS search domain.
14	Test network connectivity	`ping <host-name-or-IP-address>`	Sends ICMP Echo Requests to verify connectivity with the specified host.

Sample Verification Outputs

Verification Command	Sample Output
ip address show enp1s0	inet 192.0.2.1/24 inet6 2001:db8:1::fffe/64
ip route show default	default via 192.0.2.254 dev enp1s0 proto static
ip -6 route show default	default via 2001:db8:1::ffee dev enp1s0 proto static
cat /etc/resolv.conf	search example.com nameserver 192.0.2.200 nameserver 2001:db8:1::ffbb
ping google.com	64 bytes from google.com: icmp_seq=1 ttl=117 time=11.2 ms

RESULT: Therefore the study of various Network commands used in Linux and Windows have been simulated and verified.



File Edit View Search Terminal Help

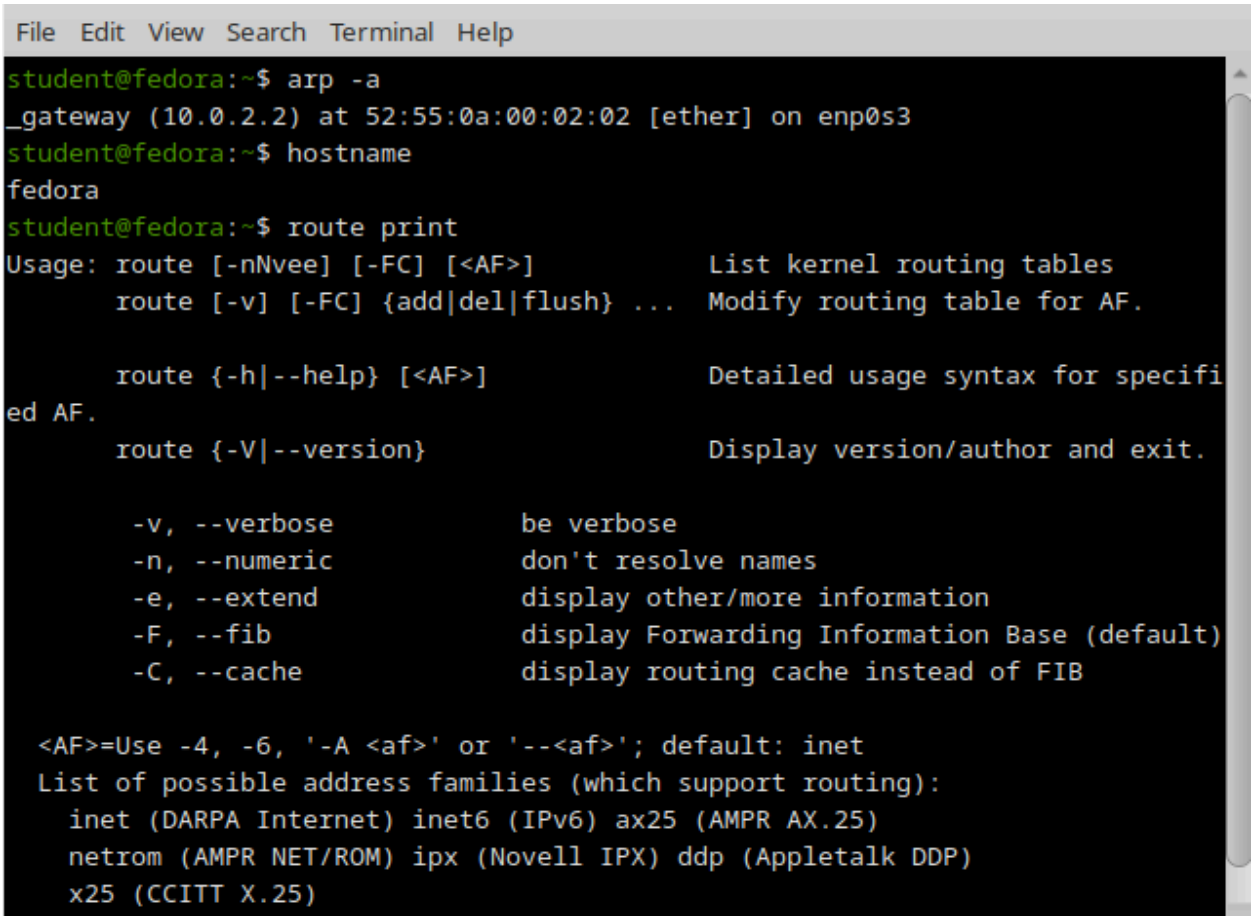
```
                         My traceroute  [v0.95]
fedora (10.0.2.15) -> google.com (142.251.220.110)   2026-07-25T09:07:39+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                   Packets               Pings
 Host                            Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. 142.251.220.110 (142.251.220.1  0.0%     4   13.0  18.0  13.0  30.0   8.0
```

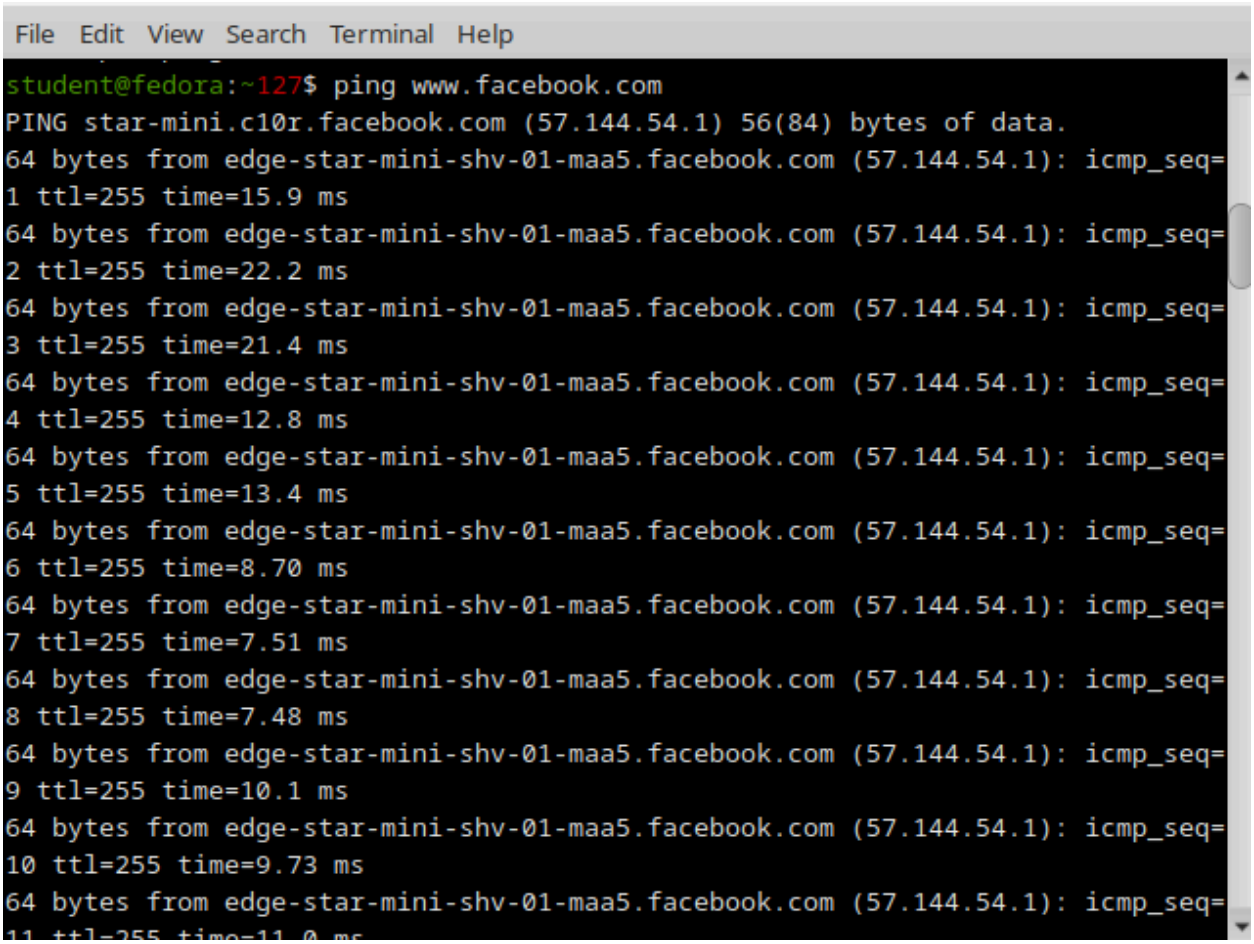
```
student@fedora:~$ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]
8.nflog (Linux netfilter log (NFLOG) interface) [none]
9.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
student@fedora:~$
```

File Edit View Search Terminal Help

```
student@fedora:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<
global>
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 138  bytes 29665 (28.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 168  bytes 17540 (17.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 20  bytes 2130 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 20  bytes 2130 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```


File Edit View Search Terminal Help
student@fedora:~$ arp -a
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3
student@fedora:~$ hostname
fedora
student@fedora:~$ route print
Usage: route [-nNvee] [-FC] [<AF>] List kernel routing tables
route [-v] [-FC] {add|del|flush} ... Modify routing table for AF.
route {-h|--help} [<AF>] Detailed usage syntax for specifi
ed AF.
route {-V|--version} Display version/author and exit.
-v, --verbose be verbose
-n, --numeric don't resolve names
-e, --extend display other/more information
-F, --fib display Forwarding Information Base (default)
-C, --cache display routing cache instead of FIB
<AF>=Use -4, -6, '-A <af>' or '--<af>'; default: inet
List of possible address families (which support routing):
inet (DARPA Internet) inet6 (IPv6) ax25 (AMPR AX.25)
netrom (AMPR NET/ROM) ipx (Novell IPX) ddp (Appletalk DDP)
x25 (CCITT X.25)


File Edit View Search Terminal Help
student@fedora:~127$ ping www.facebook.com
PING star-mini.c10r.facebook.com (57.144.54.1) 56(84) bytes of data.
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=1 ttl=255 time=15.9 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=2 ttl=255 time=22.2 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=3 ttl=255 time=21.4 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=4 ttl=255 time=12.8 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=5 ttl=255 time=13.4 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=6 ttl=255 time=8.70 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=7 ttl=255 time=7.51 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=8 ttl=255 time=7.48 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=9 ttl=255 time=10.1 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=10 ttl=255 time=9.73 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=

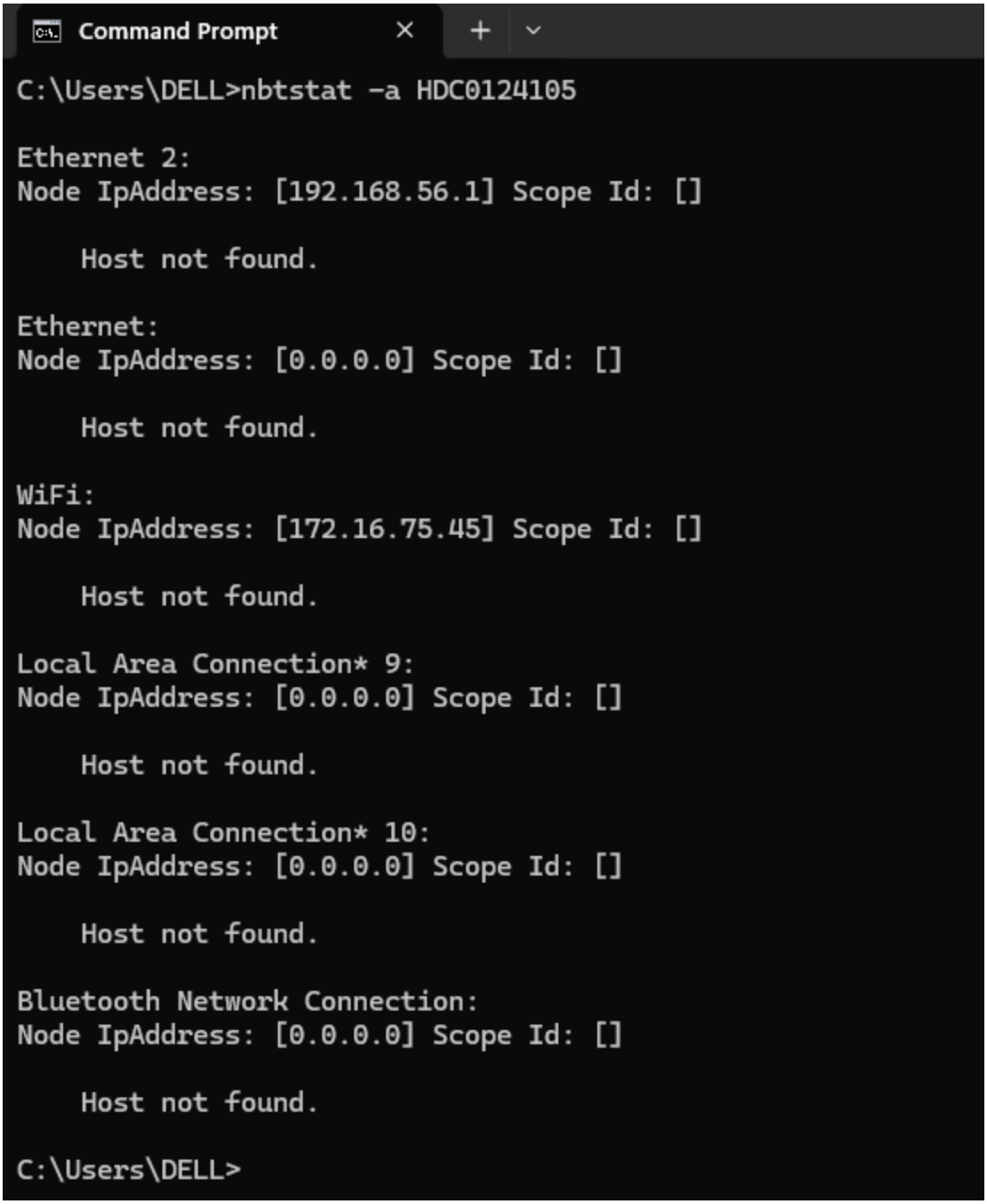
```
File  Edit  View  Search  Terminal  Help
student@fedora:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State

udp        0      0 fedora:bootpc           _gateway:bootps         ESTABLISHE
D
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     19174    /run/systemd/journa
l/stdout
unix  3      [ ]         STREAM     CONNECTED     17819    /run/systemd/journa
l/stdout
unix  3      [ ]         STREAM     CONNECTED     23930
unix  3      [ ]         STREAM     CONNECTED     22896
unix  3      [ ]         STREAM     CONNECTED     22825    /run/systemd/journa
l/stdout
unix  3      [ ]         STREAM     CONNECTED     22567    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     19354    /run/dbus/system_bu
s_socket
unix  3      [ ]         STREAM     CONNECTED     22624    /run/dbus/system_bu
s_socket
unix  3      [ ]         STREAM     CONNECTED     22817    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22591    /run/dbus/system_bu
s_socket
```

```
student@fedora:~$ netstat -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
default         _gateway        0.0.0.0         UG        0 0          0 enp0s
3
10.0.2.0        0.0.0.0         255.255.255.0   U         0 0          0 enp0s
3
student@fedora:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.77.110
Name:   google.com
Address: 2404:6800:4007:802::200e

student@fedora:~$
```


Command Prompt
C:\Users\DELL>nbtstat -a HDC0124105
Ethernet 2:
Node IpAddress: [192.168.56.1] Scope Id: []
Host not found.
Ethernet:
Node IpAddress: [0.0.0.0] Scope Id: []
Host not found.
WiFi:
Node IpAddress: [172.16.75.45] Scope Id: []
Host not found.
Local Area Connection* 9:
Node IpAddress: [0.0.0.0] Scope Id: []
Host not found.
Local Area Connection* 10:
Node IpAddress: [0.0.0.0] Scope Id: []
Host not found.
Bluetooth Network Connection:
Node IpAddress: [0.0.0.0] Scope Id: []
Host not found.
C:\Users\DELL>

```
C:\Users\DELL>pathping HDC0124105

Tracing route to HDC0124105 [172.16.75.45]
over a maximum of 30 hops:
  0  HDC0124105 [172.16.75.45]
  1  HDC0124105 [172.16.75.45]

Computing statistics for 25 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           HDC0124105 [172.16.75.45]
                                    0/ 100 =  0%   |
  1    0ms     0/ 100 =  0%     0/ 100 =  0%  HDC0124105 [172.16.75.45]

Trace complete.

C:\Users\DELL>
```

```
student@fedora:~127$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defa
ult qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83835sec preferred_lft 83835sec
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dynamic noprefi
xroute
       valid_lft 86135sec preferred_lft 14135sec
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
student@fedora:~$
```

```
File  Edit  View  Search  Terminal  Help
student@fedora:~$ mtr -g google.com
mtr: invalid option -- 'g'

Usage:
 mtr [options] hostname

 -F, --filename FILE        read hostname(s) from a file
 -4                         use IPv4 only
 -6                         use IPv6 only
 -u, --udp                  use UDP instead of ICMP echo
 -T, --tcp                  use TCP instead of ICMP echo
 -I, --interface NAME       use named network interface
 -a, --address ADDRESS      bind the outgoing socket to ADDRESS
 -f, --first-ttl NUMBER     set what TTL to start
 -m, --max-ttl NUMBER       maximum number of hops
 -U, --max-unknown NUMBER   maximum unknown host
 -P, --port PORT            target port number for TCP, SCTP, or UDP
 -L, --localport LOCALPORT  source port number for UDP
 -s, --psize PACKETSIZE     set the packet size used for probing
 -B, --bitpattern NUMBER    set bit pattern to use in payload
 -i, --interval SECONDS     ICMP echo request interval
 -G, --gracetime SECONDS    number of seconds to wait for responses
 -Q, --tos NUMBER           type of service field in IP header
 -e, --mpls                 display information from ICMP extensions
```

```
File  Edit  View  Search  Terminal  Help
                          My traceroute  [v0.95]
fedora (10.0.2.15) -> google.com (142.250.182.78)    2026-07-25T09:09:18+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                 Packets                Pings
 Host                          Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. lcmaaa-ax-in-f14.1e100.net  0.0%     3    8.1   8.8   8.1   9.7   0.8
```

```
student@fedora:~1$ nmcli connection show
NAME                UUID                                  TYPE      DEVICE
Wired connection 1  86ead386-e6ab-359e-9d82-71491148899f  ethernet  enp0s3
lo                  d3bc006d-3b51-4d5e-ba62-b097234a30ab  loopback  lo
student@fedora:~$
```

```
student@fedora:~$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolve
d(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but onl
y
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
```